# INTRODUCTION TO SYSTEMS ANALYSIS AND DESIGN:

## AN AGILE, ITERATIVE APPROACH

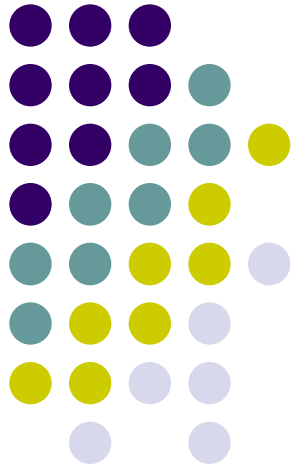
SATZINGER | JACKSON | BURD

CHAPTER 7

# Designing the User and System Interfaces

# Chapter 7

Introduction to Systems Analysis and Design:
An Agile, Iteractive Approach
6th Ed
Satzinger, Jackson & Burd



# Chapter 7 Outline

- User and System Interfaces（用户界面和系统界面）
- Understanding the User Interface（理解用户界面）
- User-Interface Design Concepts（用户界面的设计概念）
- The Transition from Analysis to User-Interface Design（从分析到用户界面设计的转换）



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# Chapter 7 Outline

- User-Interface Design（用户界面设计）
- Identifying System Interfaces（确定系统界面）
- Designing System Inputs（设计系统输入）
- Designing System Outputs（设计系统输出）



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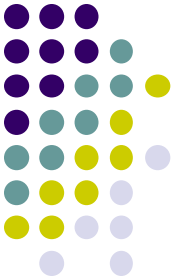
# Learning Objectives

- Describe the difference between user interfaces（用户界面） and system interfaces（系统界面）
- Describe the historical development of the field of human-computer interaction (人机交互领域的发展)
- Discuss how visibility（可视性） and affordance（提示性） affect usability
- Describe user-interface guidelines（用户界面的指导规则） that apply to all types of user-interface types and additional guidelines specific to Web pages and mobile applications



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# Learning Objectives (continued)



- Create storyboards（创建故事脚本） to show the sequence of forms used in a dialog
- Discuss examples of system interfaces found in information systems（讨论系统接口）
- Define system inputs and outputs（定义输入输出） based on the requirements of the application program
- Design printed and on-screen reports appropriate for recipients（设计适合收件人的报告）



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# Overview



- Information systems interact with people and other systems（信息系统与人和其他系统相互作用）
- Poorly designed user interface can make the information system unusable（设计不良用户界面会使信息系统无法使用）
- Poorly designed system interfaces are a source of errors and inefficiency（设计不良的系统接口是错误和低效率的根源）
- User and system interfaces involve both inputs and outputs and involve a large number of stakeholders（用户和系统接口涉及输入和输出，并涉及大量利益相关者）



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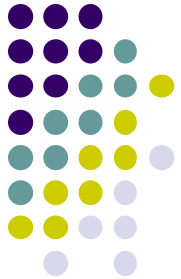
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# User Interfaces and System Interfaces



- User Interface （用户界面）– inputs and outputs that directly involve a human user/actor（直接涉及人类用户的输入输出）
- System Interface（系统接口） – the inputs and outputs that require minimal human intervention（输入和输出需要最少的人为干预）
  - Inputs captured automatically（自动捕获输入）
  - Outputs direct to other systems （直接输出到其他系统）
  - Printed and distributed outputs (打印和分发输出)



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# User Centered Design

- Design techniques that embody the view that the user interface is the system to the user（从用户的角度看，用户界面就是整个系统。界面后的程序、数据库、硬件都是不相关的）
- Principles of User Centered Design（三个重要原则）
  - Focus early on users and their work（及早关注用户及其工作）
  - Evaluate designs to ensure usability（评价设计以确保其可用性）
  - Use iterative development（使用迭代开发方法）



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# User Interface Design Concepts

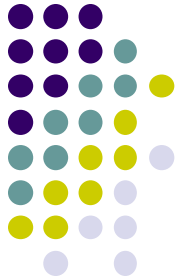
- Human-Computer Interaction (HCI)人机交互
  - A field of study concerned with the efficiency and effectiveness of user interaction with computer systems, human-oriented input and output technology, and psychological aspects of user interfaces
- Visibility and Affordance可视性和提示性
  - To be usable, a control must be visible to the user and its appearance should suggest its functionality






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# User Interface Design Guidelines



- Design for Consistency（一致性）
- Provide Shortcuts（快捷方式）
- Provide Feedback（反馈）
- Dialogs Should Yield Closure（完整的对话）
- Error Handling that Provides Guidance（错误处理）
- Easy Reversal of Actions（撤销）
- Reduce Short Term Memory Load（减轻短期记忆负担）



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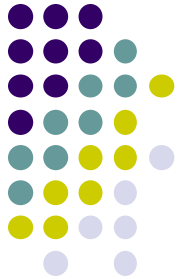
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# Use Cases and the Menu Hierarchy

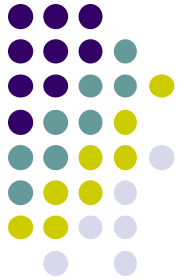


- Menus are a typical way to organize access to use case functionality（菜单是组织对用例功能访问的典型方式）
- Different types of users might have different menus（不同类型的用户可能有不同的菜单）
- Useful to design an overall menu hierarchy and then subsets for different users（可用于设计整体菜单层次结构，然后为不同用户设计子集）



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# Guidelines for Windows and Forms

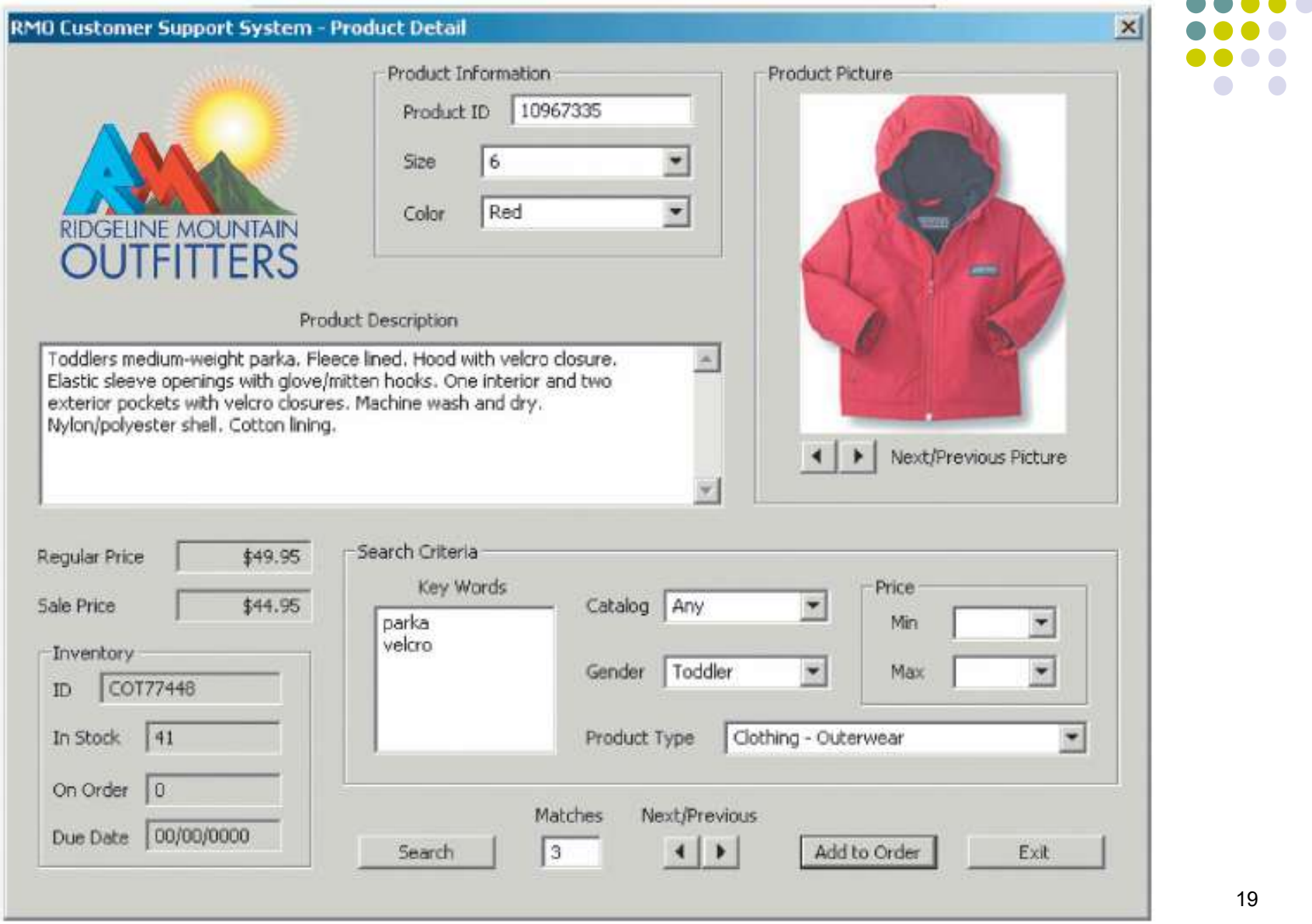


- Interface Layout and Formatting（设计窗体和表格）
  - Consistency, labels and headings, distribution and order, fonts and colors（一致性、标签和标题、分布和顺序、字体与颜色）
- Data Entry（数据输入）
  - Text box, list box, combo box, radio buttons, check boxes（文本框、列表框、组合框、单选按钮、复选框）
- Navigation and Support Controls（导航）
  - Minimize, maximize, close, scroll bars, resize（最大最小化、关闭、滚动条、调整大小）



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# RMO Windows Form





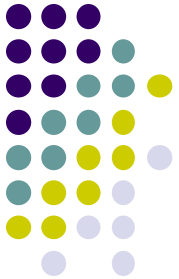

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# Guidelines for Web Browser User Interfaces

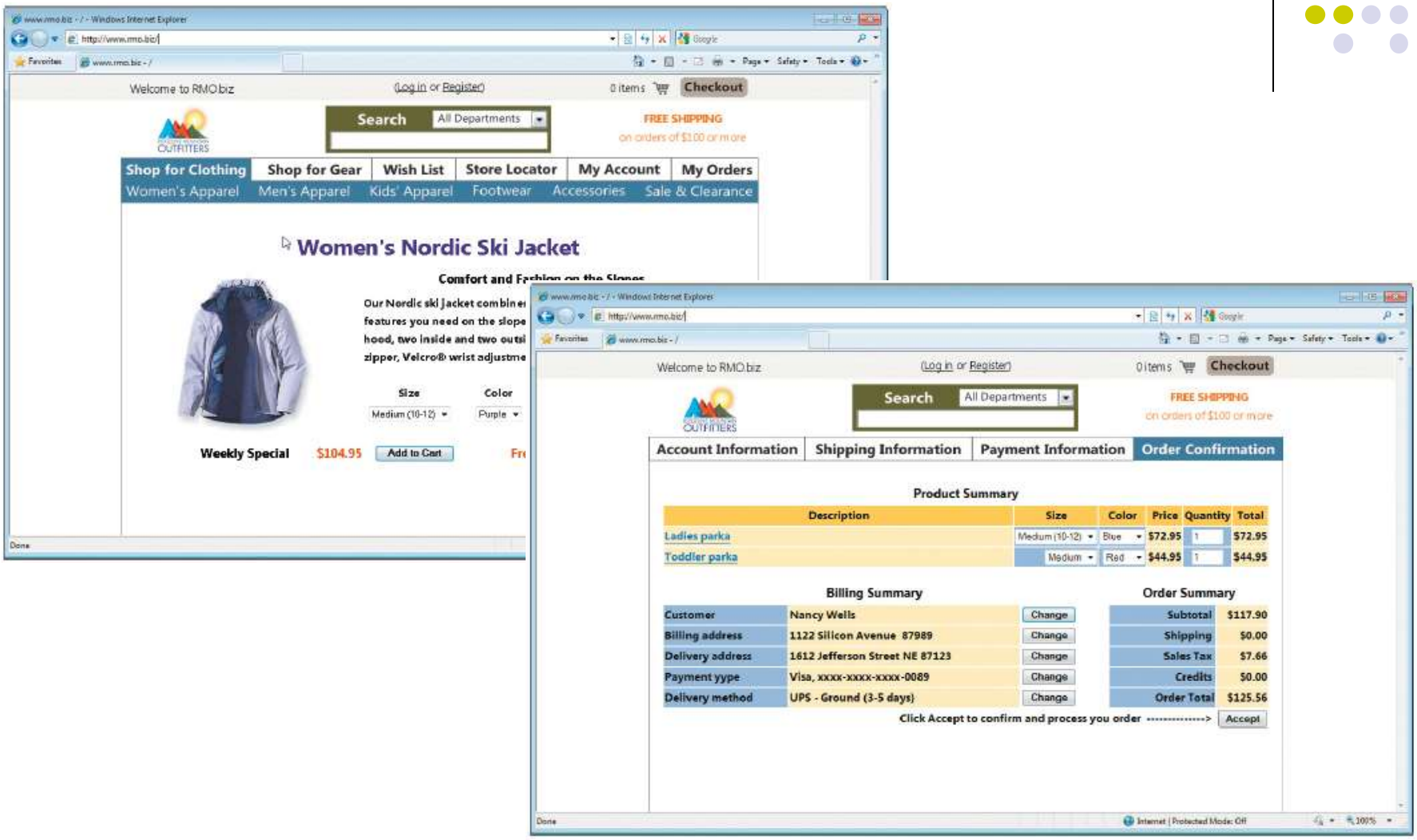


- Consistency（一致性）
  - Cascading Style Sheets (CSS) – Web page encoding standard that enables a Web site designer to specify parts of a page that will always look the same and parts that will vary by task or audience
- Performance Considerations（性能考虑）
  - Sensitive to network connection, amount of information transmitted, type of information transmitted
- Pictures, Video, and Sound（图片、视频和声音）
  - Powerful, but compatibility issues arise
- Users with Disabilities（残疾用户）
  - Assistive technologies -- software (such as text-to-speech and voice-recognition utilities) that adapts user interfaces to the special needs of persons with disabilities



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# RMO Example
## Using CSS for Consistency

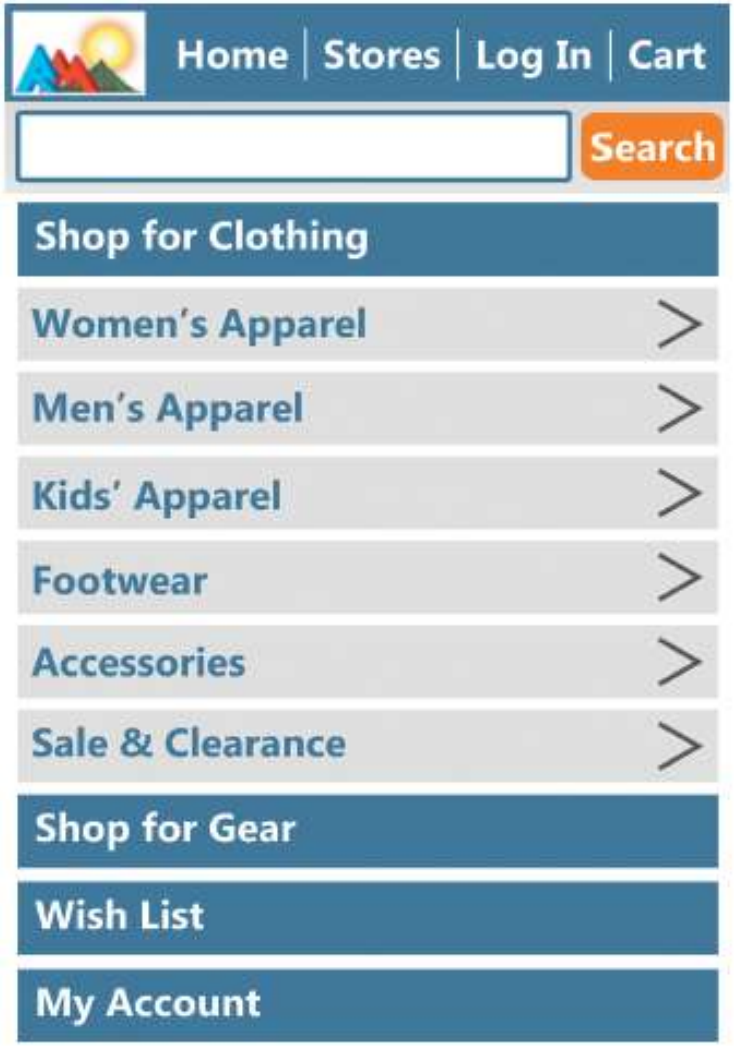





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# Guidelines for Handheld Devices（手持设备附加原则）

- Challenges
  - Small screen size, small keyboards and touch screens, limited network capacity, app design guidelines and toolkits（小屏幕、小键盘、触摸屏、有限的网络容量）






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# 人机界面设计指南

① **一般交互指南：**涉及信息显示、数据输入和系统整体控制

(1) 保持一致性。

(2) 提供有意义的反馈。

(3) 在执行有较大破坏性的动作之前要求用户确认。

(4) 允许取消绝大多数操作。

(5) 减少在两次操作之间必须记忆的信息量。

(6) 提高对话、移动和思考的效率。

① 允许犯错误。

② 按功能对动作分类，并据此设计屏幕布局。

(1) 提供对用户工作内容敏感的帮助设施

(2) 用简单动词或动词短语作为命令名。

# 人机界面设计指南

② **信息显示指南：**多种不同方式“显示”信息：用文字、图形和声音；按位置、移动和大小；使用颜色、分辨率和省略。

(1) 只显示与当前工作内容有关的信息。

(2) 不要用数据淹没用户，应该用便于用户迅速吸取信息的方式来表示数据。

(3) 使用一致的标记、标准的缩写和可预知的颜色。

(4) 允许用户保持可视化的语境。

(5) 产生有意义的出错信息。

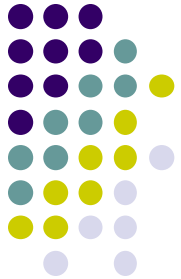
(6) 使用大小写、缩进和文本分组以帮助理解。

② 使用窗口分隔不同类型的信息。

③ 使用“模拟”显示方式表示信息，使信息更容易被用户提取。

(1) 高效率地使用显示屏。

# 人机界面设计指南



③ **数据输入指南**：用户的大部分时间用在选择命令、输入数据和向系统提供输入。

(1) 尽量减少用户的输入动作。

(2) 保持信息显示和数据输入之间的一致性。

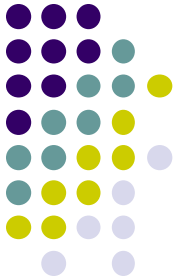
(3) 允许用户自定义输入。

(4) 交互应该是灵活的，并且可调整成用户最喜欢的输入方式。

(5) 使在当前动作语境中不适用的命令不起作用。

(6) 让用户控制交互流。

# Chapter 7 Outline

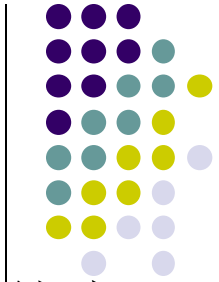


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# Identifying System Interfaces
## Inputs/outputs with minimal human intervention



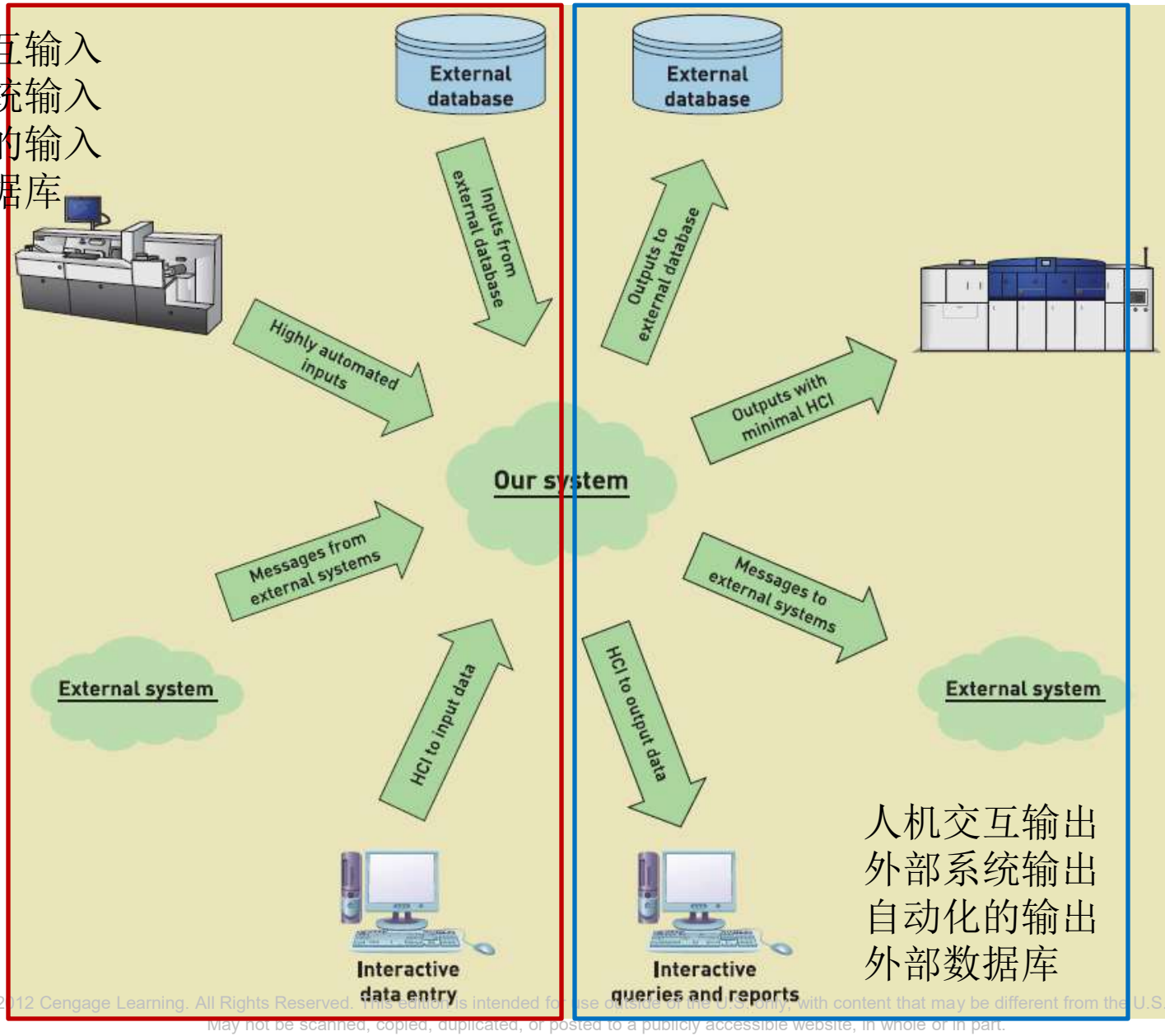
- Inputs from and outputs to other systems（与其他系统交互的输入和输出）
  - These are direct interfaces with other information systems, normally formatted as network messages.
- Highly automated inputs and outputs（高自动化的输入和输出）
  - These are captured by devices (such as scanners) or generated by persons who start a process that proceeds without further human intervention.
- Inputs and outputs to external databases（外部数据库的输入和输出）
  - These can supply input to or accept output from a system.



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# Identifying System Interfaces

人机交互输入
外部系统输入
自动化的输入
外部数据库




人机交互输出
外部系统输出
自动化的输出
外部数据库



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# System Inputs

- Primary Objective is Error Free Input（主要目标是无错误输入）
  - Use electronic devices wherever possible（尽可能使用电子设备和自动输入）
  - Avoid human involvement as much as possible（尽可能避免人工干预）
  - If information is already available in electronic form, use it instead of re-entering information（如果可以从某个电子表导出，尽可能导出而不是重新输入）
  - Validate and correct information at time and location entered（在输入信息里，对数据进行验证）



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# System Inputs

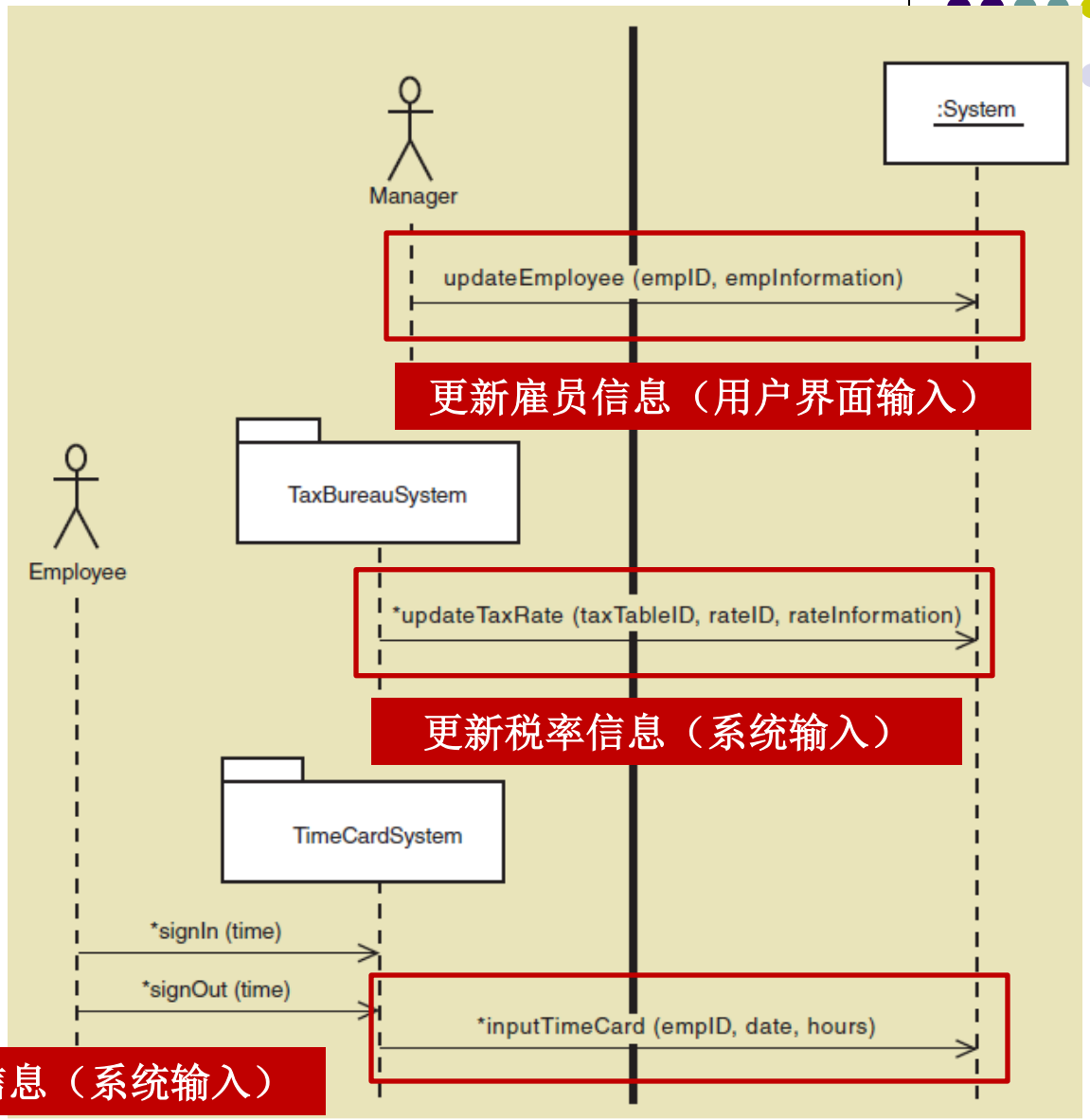


- Device Examples（设备举例）
  - Magnetic card strip readers, bar code readers, optical character recognition, radio frequency ID tags (RFID), touch screen, electronic pens, digitizers, speech recognition（磁条阅读器、条形码、光电字符识别、射频识别、触摸屏、电子币、书写板、语音识别）



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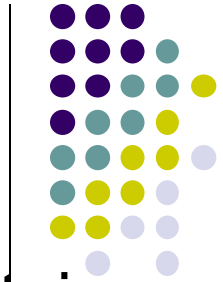
- Defining System Inputs Details
  - Sequence Diagram
  - Details for messages






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# System Outputs

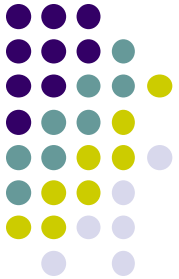


- Detailed reports（详细报告） -- reports that contain specific information on business transactions
- Summary reports（摘要报告） -- reports that summarize detail or recap periodic activity
- Exception reports （异常报告）-- reports that provide details or summary information about transactions or operating results that fall outside a predefined normal range of values
- Executive reports（执行报告） -- reports used by high level managers to assess overall organizational health and performance



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# System Outputs



- Internal outputs（内部输出） -- reports or other outputs produced for use within the organization
  - Types of internal reports apply here（为组织或单位内部使用而生成的，如报表）
- External outputs （外部输出）-- reports or other outputs produced for use by people outside the organization（为组织外部人员使用而生成的，如声明，通知，其他文档）
- Turnaround documents（返回文档） -- external outputs that includes one or more parts intended to be returned with new data or information（提供给用户的文档中包含了可能返回的一部分，如账单）



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# External Output Example



RIDGELINE MOUNTAIN OUTFITTERS

## Ridgeline Mountain Outfitters—Shopping Cart Order

**Customer Name:** Fred Westing
**Customer Number:** 6747222

**Order Number:** 4673064
**Today's Date:** May 18, 2013

**Shipping Address:**

936 N Swivel Street
Hillville, Ohio 59222

**Billing Address:**

936 N Swivel Street
Hillville, Ohio 59222

| Qty | Product ID | Description | Size | Color | Price | Extended Price |
|---|---|---|---|---|---|---|
| 1 | 458238WL | Jordan Men's Jumpman Team J | 12 | White/ Light Blue | $119.99 | $119.99 |
| 1 | 347827OP | Woolrich Men's Backpacker Shirt | XL | Oatmeal Plaid | $41.99 | $41.99 |
| 2 | 8759425SH | Nike D.R.I. – Fit Shirt | M | Black | $30.00 | $60.00 |
| 1 | 5858642OR | Puma Hiking Shorts | L | Tan | $15.00 | $15.00 |
| | | | | | | |

| | |
|---|---|
| Subtotal | $236.98 |
| Shipping | $8.50 |
| Tax | $11.25 |
| Total | $256.73 |

**Shipping Information:**

| | |
|---|---|
| Shipping Method: | Normal 7–10 day |
| Shipping Company: | UPS |
| Tracking Number: | To be sent via email |
| Email Address: | FredW253@aol.com |

**Payment Information:**

American Express ☐ MasterCard ☐ VISA ☒ Discover ☐

Account Number: X X X X – X X X X – X X X X – 5 7 8 4

MO YR

Expiration Date 05 / 15

**Thank you for your order. It is a pleasure to serve you.**
**Check back next week for new weekly specials!!**



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# Internal Output Example

**Detailed control break report**
**（控制终端详细报告）**

**Ridgeline Mountain Outfitters — Products and Items**

| ID | Season | Category | Supplier | Unit Price | Special Price | Discontinued |
|---|---|---|---|---|---|---|
| RMO12587 | Spr/Fall | Mens C | 8201 | $39.00 | $34.95 | No |

**Description** Outdoor Nylon Jacket with Lining

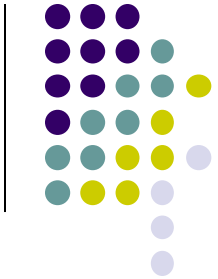
| Size | Color | Style | Units in Stock | Reorder Level | Units on Order |
|---|---|---|---|---|---|
| Small | Blue | | 691 | 150 | |
| | Green | | 723 | 150 | |
| | Red | | 569 | 150 | |
| | Yellow | | 827 | 150 | |
| Medium | Blue | | 722 | 150 | |
| | Green | | 756 | 150 | |
| | Red | | 698 | 150 | |
| | Yellow | | 590 | 150 | |
| Large | Blue | | 1289 | 150 | |
| | Green | | 1455 | 150 | |
| | Red | | 1329 | 150 | |
| | Yellow | | 1370 | 150 | |
| Xlarge | Blue | | 1498 | 150 | |
| | Green | | 1248 | 150 | |
| | Red | | 1266 | 150 | |
| | Yellow | | 1322 | 150 | |

| ID | Season | Category | Supplier | Unit Price | Special Price | Discontinued |
|---|---|---|---|---|---|---|
| RMO28497 | All | Footwe | 7993 | $49.95 | $44.89 | No |

**Description** Hiking Walkers with Patterned Tread Durable Uppers

| Size | Color | Style | Units in Stock | Reorder Level | Units on Order |
|---|---|---|---|---|---|
| 7 | Brown | | 389 | 100 | |
| | Tan | | 422 | 100 | |
| 8 | Brown | | 597 | 100 | |
| | Tan | | 521 | 100 | |
| 9 | Brown | | 633 | 100 | |
| | Tan | | 654 | 100 | |
| 10 | Brown | | 836 | 100 | |
| | Tan | | 954 | 100 | |
| 11 | Brown | | 862 | 100 | |
| | Tan | | 792 | 100 | |
| 12 | Brown | | 754 | 100 | |
| | Tan | | 788 | 100 | |
| 13 | Brown | | 830 | 100 | |
| | Tan | | 921 | 100 | |

# Drill Down Online Report
## Summary and Detailed



**Monthly Sales Summary**

Year 2013 *Month* January

| *Category* | *Season Code* | *Web Sales* | *Telephone Sales* | *Mail Sales* | *Total Sales* |
|---|---|---|---|---|---|
| Footwear | All | $ 289,323 | $ 1,347,878 | $ 540,883 | $ 2,178,084 |
| Men's Clothing | Spring | $ 1,768,454 | $ 2,879,243 | $ 437,874 | $ 4,691,484 |
| | Summer | 213,938 | 387,121 | 123,590 | 724,649 |
| | Fall | 142,823 | 129,873 | 112,234 | 384,930 |
| | Winter | 2,980,489 | 6,453,896 | 675,290 | 10,109,675 |
| | All | 1,839,729 | 4,897,235 | 349,234 | 7,086,198 |
| Totals | | | 14,747,368 | $ 1,698,222 | $ 23,391,023 |
| Women's Clothing | Spring | | | | 965,610 |
| | Summer | | | | |
| | Fall | | | | |
| | Winter | | | | |
| | All | | | | |
| Totals | | | | | |

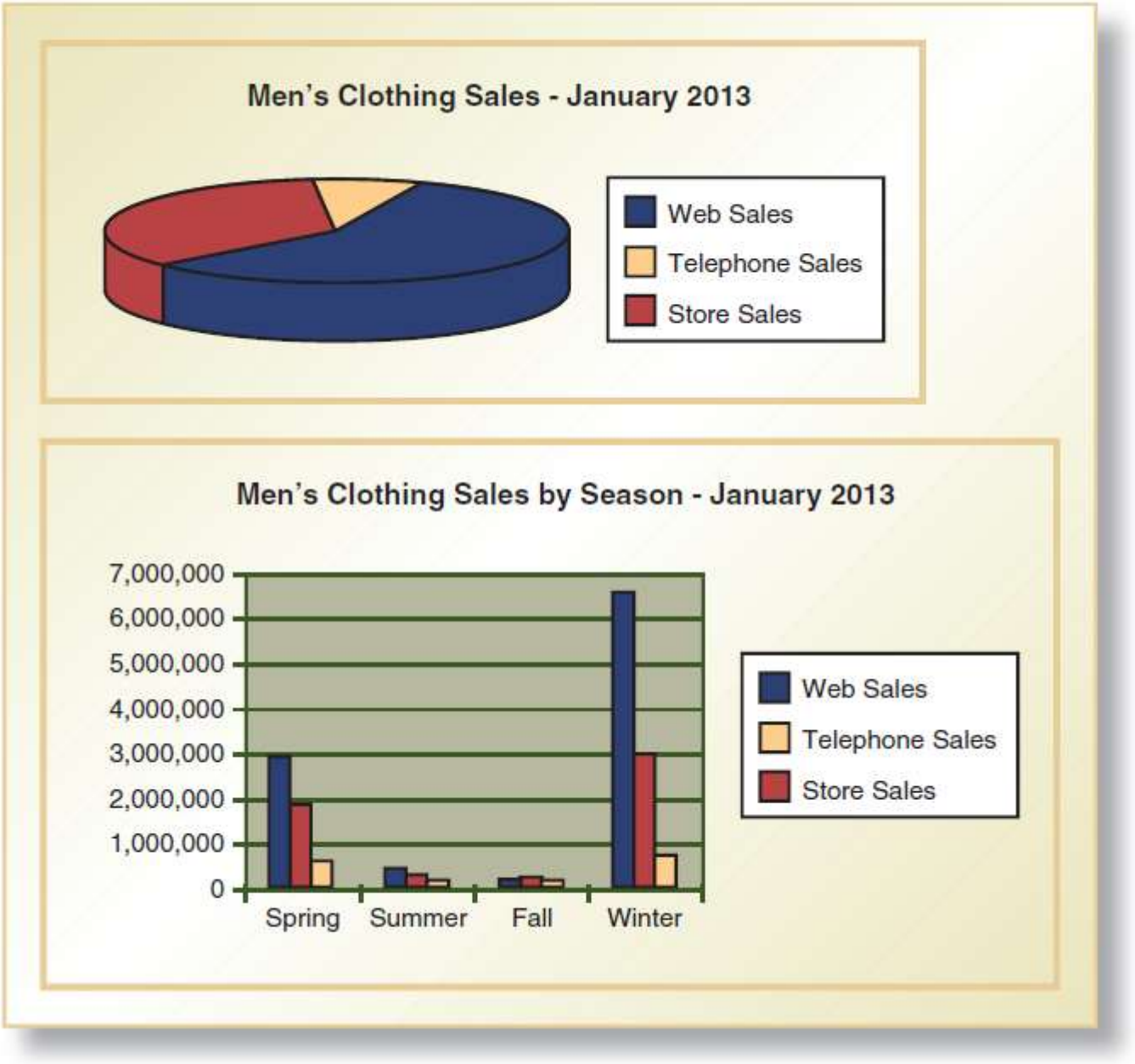
**Monthly Sales Detail**

Year 2013 *Month* January *Category* Men's Clothing *Season* Winter

| *Product ID* | *Product Description* | *Web Sales* | *Telephone Sales* | *Mail Sales* | *Total Sales* |
|---|---|---|---|---|---|
| RMO12987 | Winter Parka | $ 1,490,245 | $ 3,226,948 | $ 337,640 | $ 5,054,833 |
| RMO13788 | Fur-Lined Gloves | 149,022 | 322,695 | 33,765 | 505,482 |
| RMO23788 | Wool Sweater | 596,097 | 1,290,775 | 135,058 | 2,021,930 |
| RMO12980 | Long Underwear | 298,050 | 645,339 | 68,556 | 1,003,005 |
| RMO32998 | Fleece-Lined Jacket | 447,075 | 1,258,079 | 100,271 | 1,805,425 |
| Total | | $ 2,980,489 | $ 6,743,836 | $ 675,290 | $ 10,394,615 |

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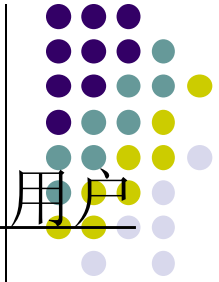
# Graphical Outputs






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# Summary

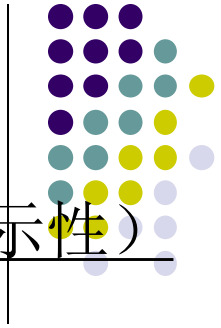


- There are two types of interfaces – user interfaces（用户接口） and system interfaces （系统接口）
- User interfaces involve direct user interaction with the system（用户与系统的直接交互）. System interfaces require minimal or no user interaction
- The design of the user interface has a long history as human computer interaction (HCI) and relies on user-centered design, which focuses early on users, evaluates designs to ensure usability, and uses iterative development（用户界面的设计作为人机交互(HCI)有着悠久的历史，它依赖于以用户为中心的设计）



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# Summary (continued)

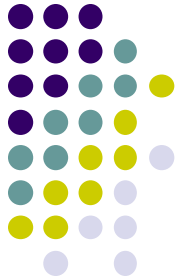


- Key user interface concepts include affordance（提示性） and visibility（可见性） for controls
- Other key principles include consistency, shortcuts, feedback, dialog closure, error handling, reversal of actions, and reducing short term memory loads（其他准则：一致性、快捷方式、反馈、对话框关闭、错误处理、操作反转和减少短期记忆负荷）
- Use cases are organized into one or more menu hierarchies to arrange functionality for users（窗体设计原则：界面布局，数据输入，导航与支持控件）
- Guidelines are available for designing for Windows, Web browsers, and Handheld devices（手持设备：小屏幕，小键盘和触摸屏，有限的网络容量）



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# Summary (continued)

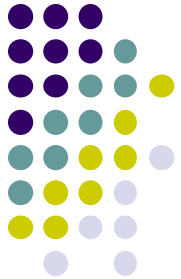


- System interfaces include inputs and outputs to other systems, highly automated inputs and outputs, and inputs and outputs to external databases.（系统接口包括对其他系统的输入和输出、高度自动化的输入和输出以及对外部数据库的输入和输出。）
- Designing system inputs involves identifying devices and mechanisms, identifying inputs and the data content, and determining the controls necessary（设计系统输入包括识别设备和机制，识别输入和数据内容，以及确定必要的控制）
- Designing system outputs includes designing detailed reports, summary report, exception reports, and executive reports（设计系统输出包括设计详细报告、摘要报告、异常报告和执行报告）



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# Summary (continued)



- Outputs are also classified as internal, external, or turnaround（输出也分为内部、外部或周转）
- Electronic reports and other outputs can include drill down, graphics, and multimedia（电子报告和其他输出可以包括深挖报告、图形和多媒体）



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